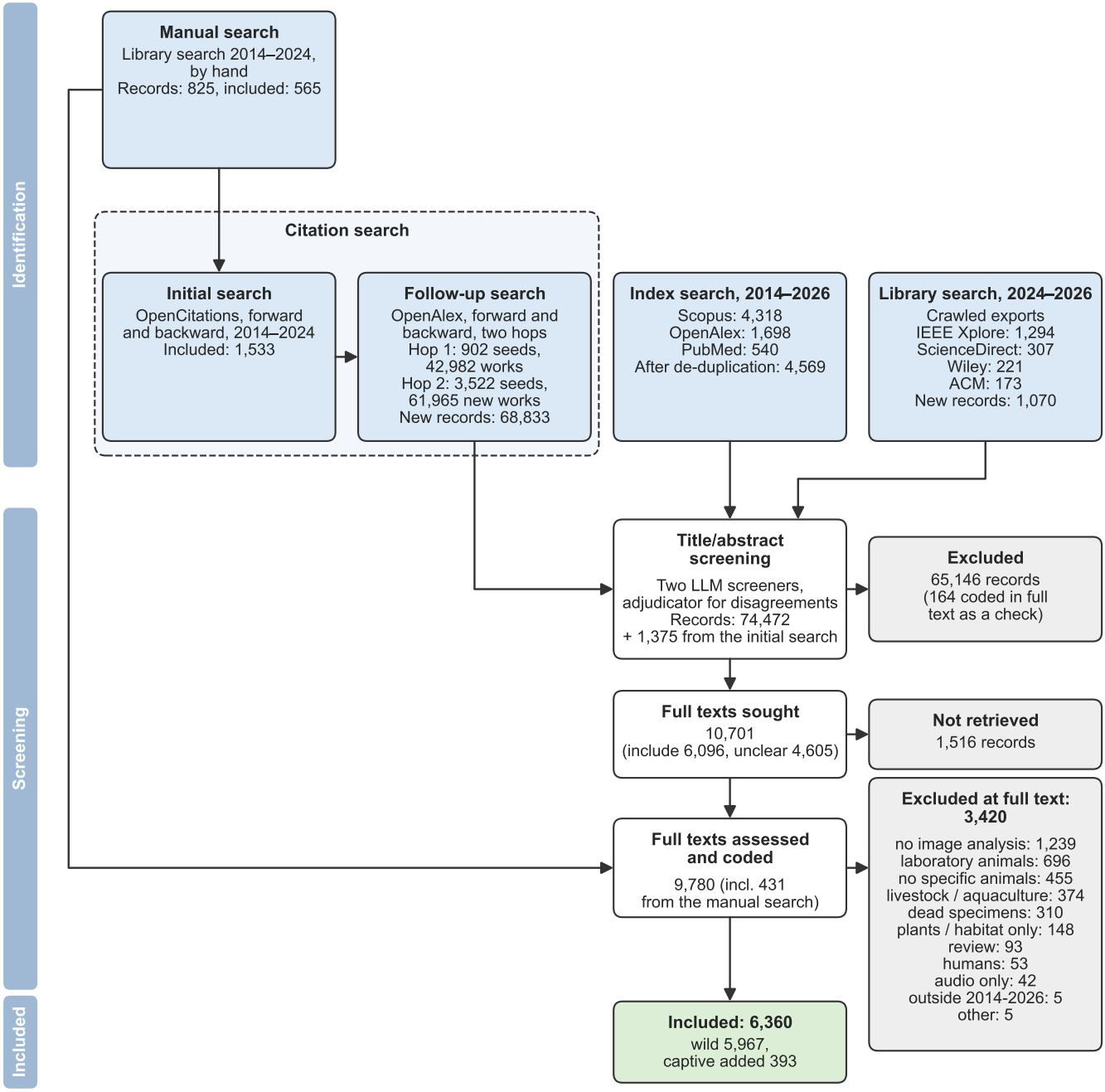
Identification
Screening
Included
Manual search
Library search 2014–2024, by hand
Records: 825, included: 565
Citation search
Initial search
OpenCitations, forward and backward, 2014–2024
Included: 1,533
Follow-up search
OpenAlex, forward and backward, two hops
Hop 1: 902 seeds, 42,982 works
Hop 2: 3,522 seeds, 61,965 new works
New records: 68,833
Index search, 2014–2026
Scopus: 4,318
OpenAlex: 1,698
PubMed: 540
After de-duplication: 4,569
Library search, 2024–2026
Crawled exports
IEEE Xplore: 1,294
ScienceDirect: 307
Wiley: 221
ACM: 173
New records: 1,070
Title/abstract screening
Two LLM screeners, adjudicator for disagreements
Records: 74,472
+ 1,375 from the initial search
Excluded
65,146 records
(164 coded in full text as a check)
Full texts sought
10,701
(include 6,096, unclear 4,605)
Not retrieved
1,516 records
Full texts assessed and coded
9,780 (incl. 431 from the manual search)
Excluded at full text: 3,420
no image analysis: 1,239
laboratory animals: 696
no specific animals: 455
livestock / aquaculture: 374
dead specimens: 310
plants / habitat only: 148
review: 93
humans: 53
audio only: 42
outside 2014-2026: 5
other: 5
Included: 6,360
wild 5,967,
captive added 393